ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)

Lab Assignment 6: Mini project using JDBC connectivity

Subject: Introduction to Databases

Subject code: CSE 3151

Programme: B.Tech. (CSE & CSIT) Semester: 6th

Course Learning Outcome	Program Outcomes	Taxonomy Level
CO2: Execute SQL queries and able to access data from a database application as per user requirements using concept of sub query and join.	PO1,PO2, PO5	L2, L3

Objective of this Assignment:

- To design a miniature project for a **University Management System** using **Java, MySQL, and JDBC**.

Requisite:

- Completion of IDB Laboratory Assignment-4
- Basic Java Programming knowledge

Overview of the Project: A University Management System is to be designed using the standard schema (Student, Instructor, Course, Takes, Teaches, Department, etc.). The project integrates a Java menu-driven program with a MySQL backend database using JDBC.

Project Description: The Java program provides an interface to perform insert, delete, update, and retrieval operations on the University database. The program handles user input, output to and from the database for the said operations. User should be able to do the following operations:

1. Show Student Records:

   Using this option the details of all the students should be displayed in particular format.

2. Add Student Record:

   Using this option the user needs to provide the information such as ID, name, Dept_name and Tot_cred through user input, which will be saved in database. After that using option 1, details of all the students will be displayed in particular format.

3. Delete Student Record:

Using this option the user needs to provide the ID of a student through user input and all the information related to that student will be deleted from the database. After that using option 1, details of all the students will be displayed in particular format.

4. Update Student Information:

   Using this option the user needs to provide the ID of a student through user input and based on the following choice the information related to the student will be updated.
   4.1 : Update name
   4.2 : Update Dept_name
   4.3 : Update Tot_cred
   After that using option 1, details of all the students will be displayed in particular format.

5. Show Instructor Details:

   Using this option the user needs to provide the ID of an instructor through user input and all the information of the instructor will be displayed in proper format.

6. Show Course Details with Enrolled Students:

   Using this option, the user provides the course_id through user input. The system displays all details of the course along with the list of students enrolled in that course (student ID, name, semester, year) in proper format.

7. Show Course Details taken by instructor:

   Using this option the user needs to provide the ID of an instructor through user input and display all the information of the course taught by instructor along with his ID in proper format.

8. Deposit HRA to Salary:

   Using this option, the user provides the instructor ID. The system calculates 15% of the current salary as HRA and adds it to the salary. The updated salary is then displayed in proper format using Option 5.

9. Deduct TDS from Salary:

   Using this option, the user provides the instructor ID. The system calculates 20% of the current salary as TDS and deducts it from the salary. The updated salary is then displayed in proper format using Option 5.

10. Exit the Program

The operations are choice based. Appropriate option has to be chosen from a switch case based menu driven program and the operation on the database is performed accordingly. The output is displayed in the terminal screen with appropriate messages from the database as displayed by MySql during direct access. Exceptions should be handled properly by the Java program. The output should be displayed in a formatted way for clarity of understanding and visual.

Program Skeleton:

```
import java.sql.*;
import java.util.Scanner;

public class jdbcproject {
        static Scanner sc = new Scanner(System.in);
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/YOUR_DATABASE_NAME";
        String user = "USER_NAME";
        String password = "PASSWORD";
        try (Connection conn = DriverManager.getConnection(url, user, password)) {
            int ch;
            if (conn != null) {
                System.out.println("Connected to the database!");
            }
            Statement stmt = conn.createStatement();
            //ResultSet rs = stmt.executeQuery("SELECT * FROM INSTRUCTOR");
            do {
                    System.out.println("\n\n***** University Management System*****");
                    System.out.println("1. Show Student Records");
                    System.out.println("2. Add Student Record");
                    System.out.println("3. Delete Student Record");
                    System.out.println("4. Update Student Information");
                    System.out.println("5. Show Instructor Details");
                    System.out.println("6. Show Course Details with Enrolled Students");
                    System.out.println("7. Show Course Details taken by Instructor");
                    System.out.println("8. Deposit HRA To Salary");
                    System.out.println("9. Deduct TDS From Salary");
                    System.out.println("10. EXIT The Program");
                    System.out.println("Enter your choice(1-10):");
                    ch = sc.nextInt();
                    switch(ch){
                            case 1:
                            // Display student records
                                    break;
                            case 2:
                            // Add student record
                            // Accept input for each column from user

                                    break;
                            case 3:
                            // Delete student record
                            // Accept student number from user

                                    break;
                            case 4:
                            // Update student record
                            // Accept student number from user
                                    System.out.println("Enter 1: For Name 2: For Dept_name  3: For
Tot_Cred to update:");
                                    // Accept user's choice switch(choice_variable_1)
                                    int ch1 = sc.nextInt();
                                    switch(ch1)
```

```
                {
                        case 1:
                                // Update student's name
                                break;
                        case 2:
                                // Update student's Dept_name
                                break;
                        case 3:
                                // Update student's Tot_cred
                                break;
                        default:
                                System.out.println("You have entered wrong choice
Please enter valid input 1,2 or 3 ");
                }
                break;
        case 5:
        // Display instructor details
        // Accept instructor ID from user break;

                 break;
        case 6:
        // Show Course Details with Enrolled Students:

        // Accept Course_id from user
        // Displays all details of the course along with the list of students

                 break;
        case 7:
        // Show Course Details taken by instructor
        // Accept instructor ID from user

        // Display all the information of the course taught by instructor

                break;
        case 8:
        // Deposit HRA to Salary

        // Accept the Instructor ID to be deposited in

        // Message for transaction completion

                break;
        case 9:
        // Deduct TDS from Salary
        // Accept the instructor ID to be withdrawn from
        // Handle appropriate withdral ckeck conditions
        // Message for transaction completion
                break;
        case 10:
        // Exit the menu
                break;
        default:
                        System.out.println("please enter valid input ");
}
```

```
        }while(ch!=10);
        conn.close();
    } catch (SQLException e) {
        System.out.println("Connection failed: " + e.getMessage());
    }
  }
}
```

Test Cases:

The program should able to produce correct answer or appropriate error message corresponding to the following testcases:

1. Show Student Records
2. Add Student Record: <12346, Swati Singh, History, 50>, < 44555, Sachin Singh, Finance, 80>, <766554, Arjun Mishra, Biology, 60>
3. Delete Student Record: <70557>, < 55739>
4. Update Student Record for any attribute except Student Number: <12346> [Update each column once]
5.
6. Show Account Details of a Student: <12345>, <00128>, <54321>
7. Show Course Details taken by instructor: <10101>, <83821>, <12121>, <76766>
8. Deposit Money to an Account: <10101>, <12121>
9. Withdraw Money from an Account: <10101>, <12121>, <76766>
10. Exit the Program
11. Enter choice 10